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An Intelligent Assistive System for Autistic Learners

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ABSTRACT

This study presents the design and implementation of an educational assistive system tailored for individuals with Autism Spectrum Disorder (ASD), leveraging eye-tracking technology and machine learning algorithms. The objective is to support communication and learning experiences through ocular gaze and fixation-based keyboard selection. An existing collection of scanpath images obtained from 15 children with ASD was utilized, and a customized heterogeneous dataset was developed by extracting fixation and gaze-related features using machine learning-based processing techniques. In total, 320 scanpath images were analyzed to capture fixation points and gaze directions during left–right and up–down navigation tasks. Based on fixation metrics and Area of Interest (AOI) analysis, a novel keyboard selection method was proposed, distinct from traditional digital keyboards. The system was evaluated using an optimized ensemble algorithm, achieving an accuracy of 83.44%, with macro-averaged F1, precision, and recall of 80.43%, 81.43%, and 80.57%, respectively, outperforming the baseline models. The results indicate the system's potential to improve communication efficiency and usability, suggesting its suitability for facilitating hands-free keyboard control. This assistive system is intended to support interaction with parents, teachers, and caregivers, which may help promote social engagement for individuals with ASD. Overall, this research demonstrates the promise of integrating eye-tracking and machine learning in educational tools, while contributing a customized dataset and a novel selection method to support neurodiverse learners. The associated code, AOI layout files, and a sample of the derived dataset regarding this study are publicly available to enable independent replication of the analysis. These materials can be accessed at <https://github.com/AfrinSadi/AssistiveAI-Autism.git>.

1 | Introduction

Communication difficulties are among the most significant challenges faced by individuals with Autism Spectrum Disorder (ASD). In particular, non-verbal or minimally verbal individuals often rely heavily on assistive technologies to engage with their surroundings. In Bangladesh, these technologies remain largely inaccessible or culturally misaligned, leading to significant barriers in educational and social integration. The integration

of eye-tracking technology and machine learning into interactive platforms, such as smart boards, offers an innovative and inclusive means of communication. However, despite their potential, existing systems often lack personalization, contextual relevance, and validation through functional communication metrics.

Assistive technologies are rapidly evolving to support individuals with Autism Spectrum Disorder (ASD), particularly those

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with limited motor or verbal abilities. While traditional systems often rely on hand-based interaction, recent approaches such as HoloGaze highlight the potential of eye-gaze in augmented environments for more natural communication. However, in Bangladesh, the adoption of such technologies remains limited due to accessibility challenges and cultural misalignment, restricting educational and social inclusion for non-verbal children. Moreover, most existing gaze-based systems lack personalization and contextual relevance, underscoring the need for culturally adaptive, inclusive solutions tailored to resource-constrained settings.

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition marked by a wide range of symptoms and behaviors that vary across individuals. Common traits include challenges in social interaction, difficulties in communication, and the presence of repetitive behaviors and sensory sensitivities [1, 2]. Individuals with ASD often struggle to maintain eye contact, interpret social cues, or engage in reciprocal communication. Additionally, co-occurring conditions such as epilepsy, mood disorders, and gastrointestinal issues may further complicate management.

Diagnosis typically involves behavioral assessments such as the Autism Diagnostic Observation Schedule (ADOS) and the Autism Diagnostic Interview-Revised (ADI-R), both of which align with the criteria outlined in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) [3, 4]. Early intervention, including approaches like Applied Behavior Analysis (ABA), speech therapy, and occupational therapy, has proven effective in mitigating developmental delays and behavioral challenges [5]. Educational support tools, such as Individualized Education Plans (IEPs), are also employed to promote academic achievement and social development [6, 7].

Advancements in technology have introduced a range of assistive tools, from low-tech communication boards to sophisticated digital platforms and mobile applications. These tools offer support in communication, daily routines, and social skills development. However, many of these technologies suffer from limitations such as inadequate sample sizes in validation studies, bias in data collection, and insufficient analysis of key behavioral markers like joint attention and gaze behavior.

Machine learning-based ASD detection systems have gained increasing attention due to their ability to support early and accurate diagnosis. In this context, [8, 9] have proposed a scenario-based questionnaire derived from established ASD symptoms to systematically capture behavioral traits. The researchers have collected a dataset of 71 responses, including both neurotypical individuals and those clinically diagnosed with ASD, and have conducted correlation analysis and principal component analysis (PCA) to identify the most significant features. They have applied four machine learning classifiers—Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), and a custom Artificial Neural Network (ANN)—to classify ASD and its subtypes. Among these, the ANN algorithm has achieved the highest accuracy of 89.8%. The study has demonstrated the effectiveness of machine learning in identifying distinct symptom patterns and supporting diagnostic decisions.

Assistive technologies for individuals with Autism Spectrum Disorder (ASD), particularly those with limited speech or motor skills, have evolved significantly in recent years. Recent advancements in Computational Intelligence techniques for the early diagnosis and monitoring of Autism Spectrum Disorder (ASD) have been reviewed by [8, 9], highlighting the use of machine and deep learning across various input modalities, as well as the challenges faced and future research opportunities.

Traditional augmentative and alternative communication (AAC) systems often require near-hand interactions, which can be physically demanding for users with motor impairments. As a response, eye-tracking technologies have gained attention for their non-intrusive and low-effort interaction capabilities. More recently, researchers have begun integrating these systems with Augmented Reality (AR) platforms to enhance communication and engagement. AR-based solutions, when combined with eye-gaze tracking, offer the potential to bypass motor limitations, enabling more accessible and intuitive interfaces for nonspeaking users. This paradigm shift signifies a growing interest in multimodal and inclusive technologies aimed at addressing the diverse needs of individuals with ASD [10].

Although recent advances have introduced innovative assistive tools for ASD, many remain inaccessible, unvalidated, or poorly suited to users in Bangladesh. Eye-tracking technologies, in particular, are rarely adapted to the linguistic and cognitive needs of local users. Moreover, few studies use real gaze data to validate whether these systems can accurately support functional tasks like character selection using machine learning.

This research addresses these gaps by developing a culturally adapted, eye-gaze-based assistive communication system tailored for individuals with ASD in Bangladesh. A customized dataset was constructed from real user data through scanpath image analysis and machine learning techniques, representing a key contribution of this work. A novel fixation and Area of Interest (AOI) based keyboard selection method was proposed and evaluated with an optimized ensemble algorithm, showing higher performance compared to baseline approaches. This study provides a data-driven reference to advance inclusive educational technologies for neurodiverse learners in underrepresented communities.

2 | Related Work

This section examines previous studies to provide a detailed assessment of advancements in assistive technologies for ASD and evaluates the proposed algorithm's performance compared to existing approaches.

2.1 | Overview of Assistive Technologies for ASD

Recent years have seen significant advancements in the development of assistive technologies tailored to individuals with ASD, particularly in the fields of eye-tracking, artificial intelligence, robotics, and immersive virtual environments.

Eye-tracking systems such as the Tobii Eye Tracker 4C and SR Research EyeLink 1000 have been widely used in autism research to study gaze behavior and joint attention [11]. Tools like Tobii Studio have facilitated the detection of fixations and saccades, enabling researchers to measure response accuracy and latency in children with ASD. These systems have provided critical insights into social cognition and have been frequently employed in both diagnostic and intervention contexts.

AI-based solutions have also been instrumental in automating diagnostic processes and personalizing interventions. Machine learning algorithms have been introduced that integrate with clinical data to support early diagnosis in primary care settings [12]. Similarly, the use of digital tools to evaluate social communication by analyzing gaze patterns and facial expressions has been highlighted [13].

Mobile applications equipped with eye-tracking capabilities have shown promise in early interventions. For instance, An Android application offering customizable social feedback has been developed to enhance joint attention skills, enabling caregivers to tailor the intervention according to individual needs [14].

Robotic systems have emerged as another promising avenue. Socially Assistive Robots (SARs) such as A-Lab Android ST, CommU, and Sota provide structured, interactive environments that facilitate communication and social skills training [15]. These robots, by simulating lifelike interactions, have been shown to improve user engagement and reduce stereotypical behaviors [16].

Virtual Reality (VR) and Mixed Reality (MR) platforms have offered immersive experiences for skill development. MR systems like Microsoft's HoloLens 2 have tracked gaze and interaction patterns to assess task engagement, making them valuable tools for individualized assessments [17–19].

One notable direction in recent research has been the development of eye-gaze-controlled systems tailored for users with motor or speech limitations. For example, the system *HoloGaze* has leveraged Microsoft HoloLens 2 to create an AR-based communication platform using gaze interaction. The system allows users to select virtual targets using either pure gaze fixation or a gaze-click mechanism (combining gaze and a button press). This setup has included calibration protocols, training sessions, and a virtual letter board designed for real-time communication. The study has involved 14 participants and has shown that gaze-click interaction is more effective and preferred by users, while also highlighting calibration and usability challenges. Such efforts have demonstrated the feasibility of deploying gaze-driven communication tools in immersive environments for nonspeaking autistic individuals [10]. The *HoloGaze* system, for instance, has explored an AR-based eye-gaze interface for nonspeaking users. While promising, such systems have remained technologically intensive and less accessible in low-income settings. Furthermore, gaze-only interaction has often led to fatigue and usability issues, which must be addressed through interface adaptation. Prior work has investigated the development of optimized virtual keyboards that adapt to users' gaze patterns. One such study has proposed a gaze-based English keyboard that uses a genetic algorithm for static layout optimization and a Markov model for

dynamic key prediction dwell time reduction. Though effective, these studies have primarily targeted general SSMI users and have not considered the unique gaze behaviors and attention span patterns associated with ASD [20].

Wearable technologies have also gained traction in this domain. For example, facial expression recognition glasses provide real-time feedback to assist in interpreting social cues, while heartbeat sensors track physiological signals to infer emotional states [21]. These insights enable the customization of emotional and behavioral interventions. Although the integration of eye-tracking, machine-learning, and other assistive technologies has shown promise, current systems rarely incorporate real-time, gaze-controlled virtual keyboards for non-verbal communication. Moreover, the need for localized, culturally sensitive solutions remains largely unaddressed. In addition to technological innovations, recent discourse analysis of Facebook status updates during COVID-19 highlighted how online speech acts particularly expressive forms—reflect identity construction, sociocultural behavior, and coping strategies [22]. Such findings underscore the role of cultural and linguistic context in shaping communication patterns, which is highly relevant when designing assistive technologies for individuals with ASD in diverse communities.

Recent advancements in gaze-based interaction systems have incorporated deep learning techniques to enhance accuracy and user experience. A notable study has developed a CNN-based face detection model (ResNet34), replacing the traditional Haar Cascade approach. This substitution has significantly improved system robustness under varying lighting, head poses, and facial features. Researchers have employed geometric modeling and facial landmark detection to extract elliptical eye contours, allowing more precise gaze estimation and blink detection. The system has implemented a multi-stage virtual keyboard interface, segmented into four regions, enabling efficient character selection through gaze direction and intentional blinks. To ensure reliable blink detection across diverse users, personalized Eye Aspect Ratio (EAR) thresholds have been calculated using baseline calibration. These individualized profiles have improved the accuracy of classifying voluntary and spontaneous blinks. Furthermore, a user study involving 10 participants has demonstrated that the integration of deep learning has increased interaction accuracy from 85.2% to 92.3%. Although CNNs have introduced higher computational demands, optimizations such as model compression and OpenCV-based acceleration have maintained real-time performance at 30 FPS. This work has highlighted the advantages of modern computer vision in overcoming the limitations of earlier handcrafted methods, underscoring the growing role of deep learning in assistive communication technologies [23].

The Hide and Seek Virtual Reality System (HSVRS), to enhance gaze fixation skills in children with Autism Spectrum Disorder (ASD) system has incorporated avatar customization using face and voice manipulation technologies, enabling the avatars to resemble familiar individuals such as parents. A pilot study involving 24 children has been conducted to evaluate the system's efficacy. The study has revealed that children in the avatar-customized group have demonstrated significantly better gaze fixation and game performance compared to those in the non-customized group. The system has included modules

for eye tracking, data logging, and adaptive prompting to support individualized learning. Subjective parental feedback and objective eye-tracking data have confirmed the positive impact of the intervention. Analysis of gaze fixation across predefined regions of interest (face, body, background) has shown significantly higher attention to the face region in the customized avatar group. These findings have indicated that deep learning-based customization in VR games can effectively promote eye contact and social engagement in autistic children. However, the study has acknowledged limitations, including small sample size and short duration, suggesting the need for extended longitudinal research [24].

2.2 | Summary of Previous Studies on Assistive Technologies

Table 1 presents studies on assistive technologies for individuals with ASD that demonstrate significant advancements in engagement, communication, and social skills. Screen readers, adaptive learning systems, and LEGO-based robotics have improved cognitive abilities and academic performance. Eye-tracking systems and gaze behavior analysis have highlighted differences in participants with ASD and have enhanced communication through letter boards or virtual keyboards. Socially assistive robots and 3D graphical algorithms have proven effective in fostering social skills, despite concerns about cost and fragility. Emotion-recognition approaches that use CNN algorithms and tools such as HoloLens 2 have shown promising results in understanding and supporting ASD-related challenges, achieving accuracies of up to 81.81%. These studies have underscored the potential of innovative tools to address the unique needs of individuals with ASD.

3 | Methods

Autism Spectrum Disorder (ASD) often poses communication challenges, particularly for individuals with limited fine motor skills, making traditional writing and typing difficult. This research introduces an innovative assistive system using eye-tracking technology for hands-free keyboard control. By enabling character selection without manual input, the system aims to enhance communication and supports parents, teachers, and caregivers, offering a valuable tool for education and speech therapy for autistic individuals.

3.1 | Gaze Points

A gaze point represents the exact location on a screen where a person is looking at a given moment. It is typically measured in (x, y) pixel coordinates.

Figures 1 and 2 represent detected gaze points, showing where a subject looked during the experiment. The blue points indicate raw gaze locations captured over time, while the red points likely represent fixations, where the eye remained stable for a threshold period. The curved structure of the points suggests a structured movement pattern, possibly as the subject followed an object on

the screen. The dense regions indicate areas of high visual attention, while more sparse regions suggest rapid transitions (saccades) between fixations.

3.2 | Area of Interest

AOI, or Area of Interest in Figure 3, refers to a specific region or location within an image or visual field that is of particular interest or relevance for analysis or observation.

When calculating the Euclidean distance using Equation (1) between two points, the AOI would be the specific coordinates of those points on a 2D plane.

$$d(u, v) = \sqrt{(u_x - v_x)^2 + (u_y - v_y)^2} \quad (1)$$

The Equation (1) represents the straight-line distance between the two points, without any turns or bends. In this study each key on the keyboard has specific X and Y coordinates that define its AOI. Analyzing the user's eye gaze fixations within the AOI of each key can help determine the character they are most likely trying to select. In summary, the term AOI refers to the specific regions or areas within a visual field that are of interest for analysis or observation, such as the coordinates of points in a 2D space or the locations of keys on a keyboard.

3.3 | Keyboard Used for Mapping

Figure 4 illustrates the use of the *QWERTY* keyboard layout with coordinates to analyze the user's eye-gaze fixations. The *QWERTY* layout is a standard keyboard arrangement that is widely used in computers and other electronic devices.

For better results, assume that the screen size is $1920 * 1080$. Each key size is $50 * 50$ except the space button has a size of $100 * 50$.

3.4 | Dataset Pre-Processing

The pre-processing phase of any machine learning work is crucial to ensure the quality and consistency of the data used for modeling. This section outlines the detailed steps undertaken to pre-process and normalize a dataset consisting of 320 rows and 64 columns. The dataset primarily comprises fixation coordinates derived from eye-tracking data, which are integral to building predictive algorithms. After pre-processing, including the removal of duplicate entries, the dataset was refined to 308 rows while maintaining 64 columns.

3.4.1 | Checking for Missing Values

Missing data can negatively impact algorithm performance, leading to biased or inaccurate results. Therefore, identifying and addressing missing values is a crucial step in pre-processing to ensure data completeness and confirm that no further imputation or data removal is required. The analysis indicated that the dataset contains no missing values, with all columns and

TABLE 1 | Summary of Recent Research Work.

Methodology	Assistive technology	Dataset	Result	Reference(s)
Mixed approach: surveys, interviews, and technical assessments	Screen readers, speech recognition software, and text-to-speech applications	Diverse	Showned improvements in engagement and academic performance	[11]
Measured the heart rate, classifies the emotional state, and communicated the results to guardians, 3D graphical algorithms to help children improve their social skills and eye contact	Heartbeat sensor, Mobile application, 3D graphical Algorithms	20 autistic child	99.8%	[21]
Used a social skills training program involving multiple humanoid robots to help individuals with autism spectrum condition to improve their social skills	Multiple humanoid robots	14 individuals	All sociability index items improved	[15]
Used a low-cost eye-tracking system with image processing	Hands-free computer mouse control system	Images recorded by a webcam	Improved mouse cursor control for individuals with limited mobility	[25]
Eye tracking study comparing gaze behavior in children with ASD and TD using human face and classroom scene stimuli	NA	79 participants	Children with ASD exhibited differences in gaze behavior compared to TD children	[26]
SIT used to assess social interaction deficits in ASD	SIT	80 participants	73%	[13]
Used a repeated measures design, alternating between adaptive and traditional learning systems. Multimodal sensors tracked emotional states, and multilevel modeling assessed effects on engagement and learning outcomes	An adaptive learning system that employed multi-modal sensor data and ML algorithms	67 participants	Increased engagement, reduced boredom significantly	[27]
Employed a single case research design, combining quantitative analysis based on student assessment scores and qualitative analysis based on interviews with special education teachers	LEGO Mind storms EV3, PVBOT	8 participants	Positive impact	[28]
Focused on group discussions to explore expectations, concerns, and barriers related to the implementation of robots in education settings with students on the autism spectrum	NAO humanoid robot developed by SoftBank Robotics, MAX software platform and mobile application	Participants were recruited from Aspect Think Tank, Aspect education staff, and Australian organizations	Participants showed interest in using Socially assistive robots for teaching, despite cost and fragility concerns	[29]
16 personalized 15-min sessions for ASD participants using HoloLens 2 with eye-tracking, analyzing progress, fixation durations, and task execution times	HoloLens 2headset, MR device, ARRET toolkit and R programming language	Raw eye-tracking data	81.81%	[17]

(Continues)

TABLE 1 | (Continued)

Methodology	Assistive technology	Dataset	Result	Reference(s)
Proposed a transfer learning based CNN algorithm for emotion recognition from facial images	Sensors	Facial expression recognition datasets from Kaggle and CK + 48	Inception ResNetV2 Architecture 78.56%, Cross validation 76.70%	[30]
Used head mounted eye tracking to study non-speaking autistic individuals using letter boards, assessing their communication process	Participants used letter boards with Assistance from an assistant who held the board	9 nonspeaking autistic participants	Participants pointed to letters quickly and accurately, suggesting they conveyed their thoughts	[31]
Developed a system that detects eye gaze and eye blinking to control a virtual keyboard. Used facial landmarks, HoG features, and a threshold-based method for detection	A system	NA	91%	[32]
A literature review of 13 studies focusing on the use of SARs in interventions for children with ASD. The studies examined the impact of SARs on various aspects of social skills in children with ASD, such as eye contact, verbal communication, play skills, stereotypical behavior, engagement, and positive affect	Socially Assistive Robots (SARs)	NA	SARs can have a positive impact on the social skills and communication abilities of children with ASD	[16]
The paper primarily reviewed and discussed various AI-based approaches to ASD diagnosis and treatment, highlighting innovative technologies and methods in the field	AI-based Software as a Medical Device designed to aid in the diagnosis of ASD in primary care settings	NA	NA	[12]
Review	VR	NA	Positive improvement in social functioning, emotion recognition, and speech or language in individuals with ASD	[33]
Developed an Alexa-based skill called ASPECT to measure the recognition of utterances by autistic children. Participants inter-acted with ASPECT during therapy sessions aimed at improving vocalizations and echoic responses.	ASPECT	NA	NA	[34]

(Continues)

TABLE 1 | (Continued)

Methodology	Assistive technology	Dataset	Result	Reference(s)
A user-centered design approach, gathering requirements through interviews with psychologists, therapists, and parents of children with ASD. The researchers implemented the Scrum methodology during development	Squizzy	Data were collected through interviews with psychologists, therapists, and parents of children with ASD from YCHI Autism Cen- ter. Ten children with ASD tested Squizzy during therapy sessions, accompanied by therapists or parents	Positive Feedback	[35]
The ASDesign method, involves user analysis, context analysis, and defining needs through modules ASDT1, ASDT2, and ASDT3. It focuses on understanding the basic needs of ASD children and their specific needs during confinement, translating them into functional requirements and design parameters for assistive technology	Visual tools like picture exchange Communication systems (PECS), mobile applications and robots	Analysis of existing research, Interviews with professionals, and a Questionnaire completed by parents of children with ASD	Showed promising results	[36]
A systematic literature review conducted in three phases according to Kitchenham's method. Research questions were formulated, studies were selected based on inclusion/exclusion criteria, data extraction and synthesis were performed, and the results were analyzed to address the research questions	VR glasses, virtual assistant, eye tracker, HMD, spatial audio, and Leap Motion device	Not specific	Review paper	[18]

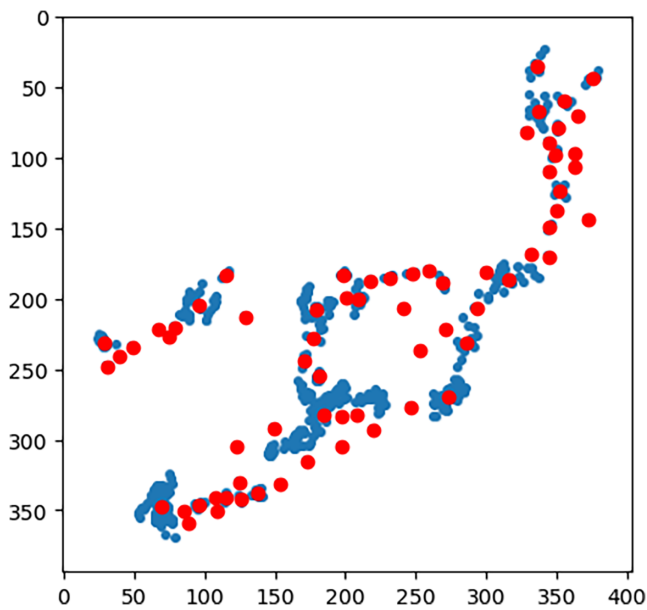


FIGURE 1 | Detected gaze points 1.

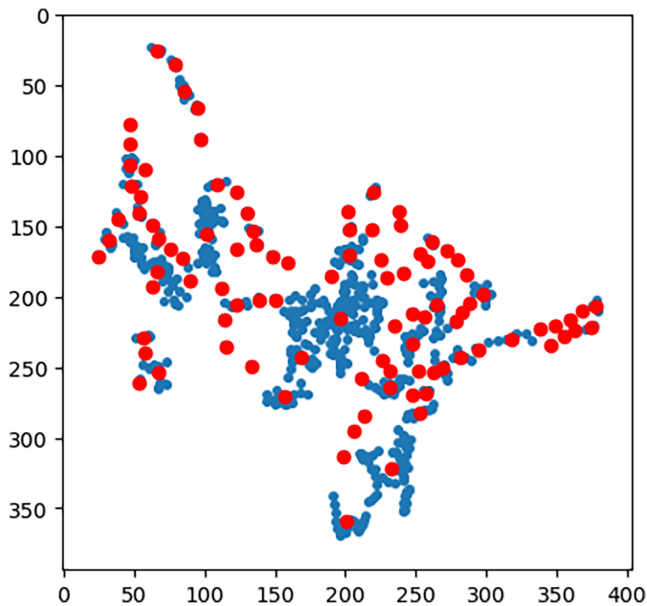


FIGURE 2 | Detected gaze points 2.

rows showing zero missing entries. Missing values were handled through *implicit row removal* and by assigning a value of zero to empty cells.

3.4.2 | Normalization of Continuous Variables

Normalization is vital for ensuring that the continuous features in the dataset are scaled appropriately, particularly when they have varying units or ranges. *Min-Max Scaling* was employed to rescale all continuous variables to a range of 0 to 1, preserving the relationships between data points while making all features non-negative to scale features to a consistent range, facilitating convergence during algorithm training, to ensure all fixation values remain non-negative, as negative values are non-physical for

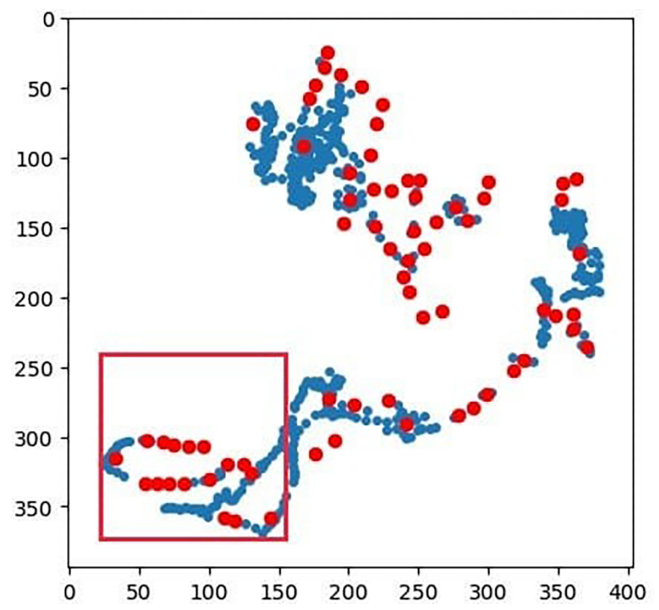


FIGURE 3 | Area of interest visualization for fixation and gaze analysis.

coordinates in this context to improve algorithm interpretability and compatibility.

3.4.3 | Final Clean-Up

During this phase, the dataset was refined further for modeling. Non-essential columns, such as 'Image name,' were removed, ensuring the dataset contains only relevant features for prediction. It's necessary to reduce dataset size and remove extraneous information that does not contribute to algorithm learning and to streamline the dataset for machine learning tasks.

3.4.4 | Validation of the Final Dataset

After pre-processing, the final dataset was inspected to ensure consistency and readiness for modeling to verify that all pre-processing steps were applied correctly and to ensure that the dataset is formatted and ready for machine learning. These involved inspecting the first few rows of the dataset to get an overview of how it looks after all the pre-processing or transformations and checking the size of the dataset (number of rows and columns), which helps confirm that no data was lost during any operations.

3.4.5 | Dataset Augmentation

Dataset augmentation is a critical step in machine learning, particularly when working with limited data. Augmentation enhances the diversity of the dataset by generating additional samples, thereby reducing overfitting and improving algorithm generalization. In this case, the dataset was augmented to expand the feature space and labels. To enrich the dataset, augmentation techniques such as adding *Gaussian Noise* and applying *small shifts* were used. These techniques simulate variations in

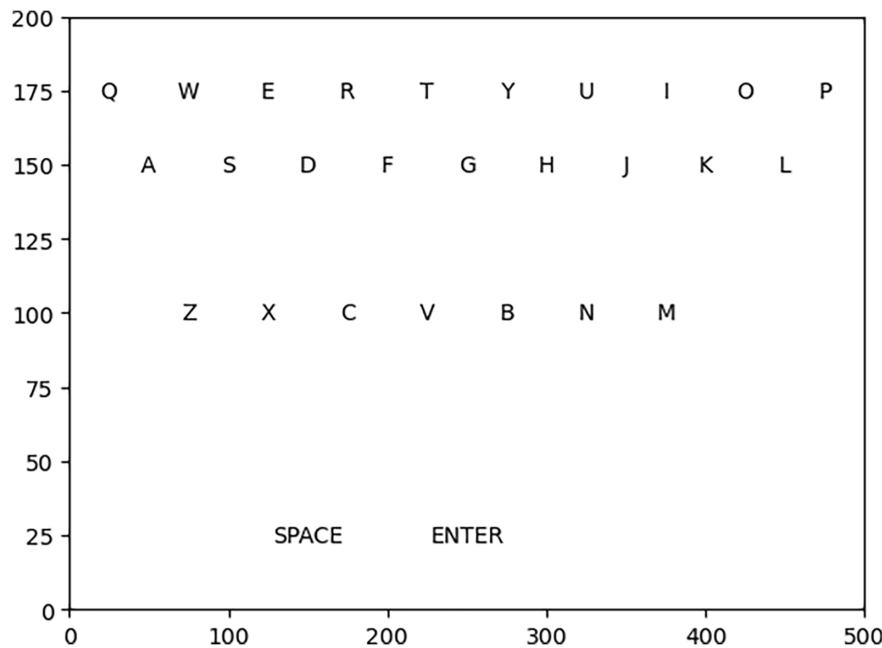


FIGURE 4 | Keyboard coordinate layout for mapping.

the data, effectively increasing its size and variability while preserving the underlying distribution. Augmentation Techniques include *Gaussian Noise* (Random noise with a mean of 0 and a standard deviation of 0.01 was added to each feature, simulating natural variability in fixation data) and *Uniform Shifts* (Small random shifts within a range of -2 to 2 were applied, representing potential minor inconsistencies in gaze tracking).

The algorithm enhanced data augmentation, custom ensemble learning, and optimized meta-learning approach, all of which contribute to its robustness and superior performance. The enhanced data augmentation technique generates diverse training samples by introducing Gaussian noise, random shifts, and scaling, helping the algorithm generalize better and handle data variations effectively.

3.4.6 | Dataset Size Verification

Post-augmentation, the size of the dataset was verified to confirm the successful addition of augmented samples. The dataset expanded from the original size to include augmented samples, with each feature vector aligned to its corresponding label to validate the effectiveness of augmentation and ensure all new data points are accounted for. The Size of X (feature) is (1540, 62) and the size of y (labels) is (1540).

3.4.7 | Handling Class Imbalance

The dataset is real and self-constructed, ensuring its authenticity and relevance for the problem at hand. However, as illustrated in Figure 5, the dataset initially exhibited significant class imbalance, with certain classes having very few samples while others were heavily overrepresented. Such an imbalanced distribution can cause machine learning algorithms to favor the majority

classes, leading to biased predictions and poor generalization. To address this, *Random Over-Sampling* was applied to the minority classes, artificially increasing their sample sizes to balance the class distribution. Additionally, the raw dataset contained duplicate entries, missing values, and inconsistencies, which could negatively impact the algorithm's performance.

To address these challenges and improve the dataset's quality, a series of pre-processing techniques were applied. First, duplicate entries were removed to ensure that redundant data points did not skew the learning process. This helped eliminate bias and made the dataset more representative. Next, missing values were handled appropriately, either through imputation techniques or by carefully removing incomplete records that could introduce noise into the training process. Additionally, normalization techniques were applied to standardize the dataset, ensuring that all feature values were on a similar scale. This step was crucial for preventing certain features from dominating the learning process due to differences in magnitude. As shown in Figure 6, the refined dataset demonstrates a noticeable improvement. The total number of samples has increased across all classes, which enhances the algorithm's ability to learn meaningful patterns. While the dataset remains imbalanced, strategic handling of class distributions ensures that underrepresented categories are still utilized effectively. By expanding the dataset and refining its structure, the algorithm can now generalize better and avoid overfitting to dominant classes. Additionally, data augmentation was implemented to ensure that the algorithm does not disregard minority classes.

Overall, these pre-processing steps have significantly improved the dataset's quality, making it more suitable for training a robust machine-learning algorithm. The refined dataset, despite its inherent imbalance, has been carefully structured to maximize learning potential, leading to improved accuracy and better real-world performance.

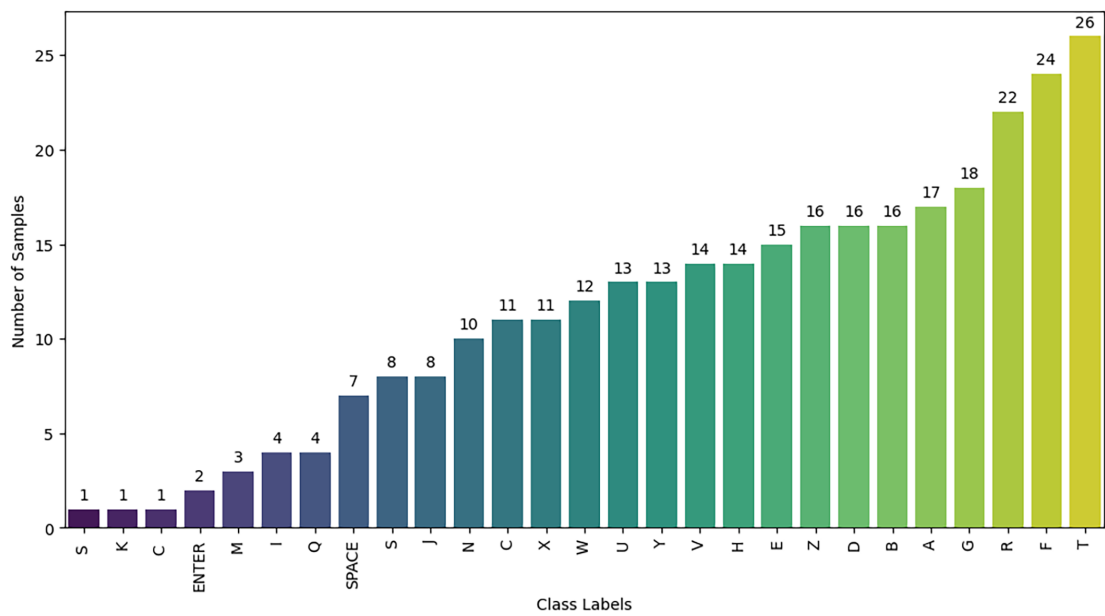


FIGURE 5 | Class distribution before dataset pre-processing.

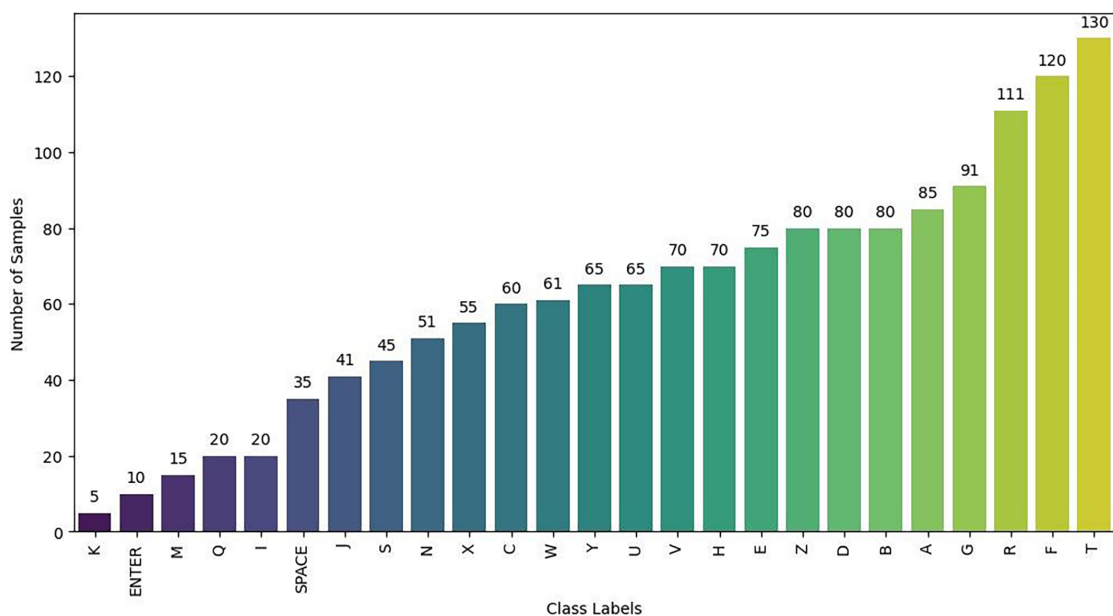


FIGURE 6 | Class distribution after dataset pre-processing.

3.4.8 | Label Encoding

To prepare the target variable for machine learning algorithms, the labels were encoded into a numeric format using the *LabelEncoder*. This step not only makes the dataset compatible with classification algorithms but also streamlines the learning process, as numerical values are easier to handle and manipulate for algorithm training and evaluation.

3.4.9 | Splitting the Dataset

The augmented dataset was systematically divided into training and testing sets to thoroughly evaluate the performance of

the machine learning algorithm. A *Stratified Shuffle Split* ratio (80–20) was employed, ensuring that a substantial portion of the data was available for effective training while preserving a representative subset for reliable and accurate testing. This approach ensured a balanced evaluation of the algorithm’s predictive capabilities.

3.5 | Dataset Creation Flowchart

To evaluate the efficacy of the system in facilitating communication through ocular gaze and fixation-based selection, a custom dataset consisting of scanpath images from 15 children diagnosed with Autism Spectrum Disorder (ASD) was created.

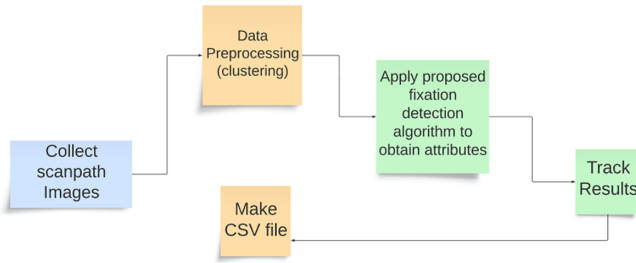


FIGURE 7 | Data collection flowchart.

The process shown in Figure 7 began with collecting scanpath images, followed by data pre-processing with clustering to extract key features and the proposed Fixation Detection Algorithm 1 was then applied to generate essential attributes. The refined data was organized into a CSV file for structured analysis.

3.6 | Determining Fixation Co-Ordinates From Scanpath Images

To enhance character prediction accuracy for hands-free keyboard control, the machine learning algorithm utilized several fixation-related features extracted from scanpath images using computer vision techniques and mathematical computations. Fixation values refer to representative spatial coordinates where a user's gaze remained stable over time, signifying focused attention on specific regions of the screen. These fixation values were essential for linking gaze behavior to character selection in a virtual keyboard interface designed for individuals with Autism Spectrum Disorder (ASD).

For the creation of the fixation-based dataset, no machine learning algorithms were used. Instead, a rule-based approach was employed, relying on classical image processing techniques and geometric heuristics. This approach involved extracting gaze points from scanpath images using contour detection and computing fixations based on spatial proximity using Euclidean distances. The method is conceptually aligned with dispersion-based fixation detection algorithms, which are particularly effective for structured and noise-free input and do not require prior training data. Consequently, the dataset generation process was deterministic and interpretable, making it highly suitable for controlled experimental studies preceding the integration of machine learning algorithms.

Fixation detection was performed using a custom image processing pipeline based on the Suzuki & Abe's (1985) *border-following algorithm*, which identified contours corresponding to gaze points [37]. The scanpath image(α) was first converted to grayscale (`cvtColor(image, COLOR_BGR2GRAY)`), followed by binary inverse thresholding (`threshold(gray, 128, 255, THRESH_BINARY_INV)`) to segment gaze path dots from the background. Using the `findContours` function with `RETR_EXTERNAL` and `CHAIN_APPROX_SIMPLE`, external contours representing potential gaze points were extracted. Each

contour was then analyzed, from which the centroid of the gaze point was calculated using Equations (2) and (3).

$$C_x = \frac{M[m01']}{M[m00']} \quad (2)$$

$$C_y = \frac{M[m10']}{M[m00']} \quad (3)$$

The centroids represented gaze points and were stored sequentially. To group these gaze points into fixations (β), a distance-based clustering method was applied. Using a threshold of 20 pixels, points were clustered based on the Equation (1) ([38]). If the distance between them was less than or equal to 20 pixels, they were considered part of the same fixation group. The representative fixation value was then computed as the average of all gaze points within a cluster using Equations (4) and (5):

$$fixation_x = \frac{1}{n} \sum_{i=1}^n x_i \quad (4)$$

$$fixation_y = \frac{1}{n} \sum_{i=1}^n y_i \quad (5)$$

where n is the number of gaze points in the fixation cluster. This feature formed a structured dataset that enabled machine learning algorithms to interpret eye-tracking data accurately for hands-free keyboard control, making communication easier for individuals with Autism Spectrum Disorder (ASD).

Algorithm 1 extracts fixation points from a scanpath image (α). Fixations (β) are determined by clustering gaze point centroids derived from image contours, based on a distance threshold. The result is a set of fixation coordinates that represent stable gaze regions. To enhance clarity, a flowchart has been included in Figure 8 to illustrate the process more effectively.

3.7 | Character Selection Process

This section presents the algorithm proposed for predicting the selected character on a visual keyboard using eye-tracking data. The method relies on fixation frequency analysis in conjunction with spatial proximity measurements within predefined Areas of Interest (AOIs). The goal is to accurately identify the character that received the highest visual attention, thereby supporting a novel keyboard selection technique for individuals with Autism Spectrum Disorder (ASD).

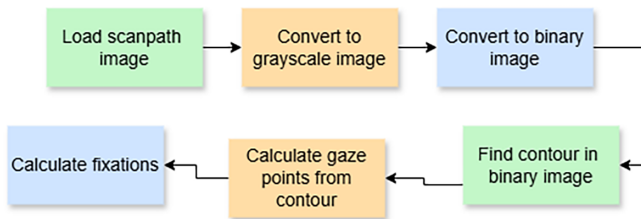
Each key on the visual keyboard interface is associated with a rectangular Area of Interest (AOI). These AOIs were predefined using bounding box coordinates specified in a layout file. Each AOI is defined by four boundaries: X_MIN , X_MAX , Y_MIN , and Y_MAX , which represent the minimum and maximum x and y coordinates encompassing the key. This layout remained consistent for all participants to ensure standardization of fixation mapping and to facilitate objective algorithm training and evaluation. Individual customization of AOIs was not

ALGORITHM 1 | Fixation detection.FIXATION-DETECTION (α, β).

```

1. gray  $\leftarrow$  Convert  $\alpha$  to grayscale using cvtColor( $\alpha$ , COLOR_BGR2GRAY)
2. THRESH_BINARY_INV  $\leftarrow$  Inverted binary thresholding constant
3. threshold_value, binary  $\leftarrow$  Apply threshold(gray, 128, 255, THRESH_BINARY_INV)
4. RETR_EXTERNAL  $\leftarrow$  Contour retrieval mode to extract only outer contours
5. CHAIN_APPROX_SIMPLE  $\leftarrow$  Contour approximation method
6. contours, hierarchy  $\leftarrow$  findContours(binary, RETR_EXTERNAL, CHAIN_APPROX_SIMPLE)
7. gaze_points  $\leftarrow$  empty list
8. for each contour in contours
9.   do  $M \leftarrow$  moments(contour) //  $m00, m10$ , and  $m01$  are spatial moments of the contour
10.  if  $M[m00'] \neq 0$  then
11.     $C_x \leftarrow \frac{M[m10']}{M[m00']}$  // x-coordinate of centroid
12.     $C_y \leftarrow \frac{M[m01']}{M[m00']}$  // y-coordinate of centroid
13.    Append  $(C_x, C_y)$  to gaze_points
14. fixation_threshold  $\leftarrow$  20 // Distance threshold for grouping fixations
15.  $\beta \leftarrow$  empty list
16. if length of gaze_points  $> 0$ 
17.   then current_fixation  $\leftarrow$  [gaze_points[0]]
18.   for point from gaze_points[1] to gaze_points[end]
19.     do if euclidean_distance(point, current_fixation[-1])  $\leq$  fixation_threshold
20.       then append point to current_fixation
21.     else
22.       fixation  $\leftarrow$  average(current_fixation)
23.       Append fixation to  $\beta$ 
24.       current_fixation  $\leftarrow$  [point]
25.   fixation  $\leftarrow$  average(current_fixation)
26.   Append fixation to  $\beta$ 
27. return  $\beta$ 

```

**FIGURE 8** | Fixation calculating flowchart.

applied, as the focus of the study was to assess the algorithm's general performance across participants under a uniform interface configuration.

To determine whether a given fixation point (X, Y) lies within an AOI, the algorithm evaluates a condition. This check is implemented via a Boolean function *is_within_aoi()* and is applied to each fixation point in relation to each character's bounding box. The algorithm iterates over each fixation point and checks whether it lies within the AOI of a given character. If a fixation point falls within the AOI, a counter variable *fixation_frequency* associated with that character is incremented. This step is repeated for all fixation points and all characters.

After processing all fixation points, the character(s) with the highest fixation frequency is identified. This step represents the

initial filtering phase, isolating the subset of keys that received the most visual attention based on the number of within-AOI fixations.

To further refine the selection, the algorithm computes the *average Euclidean distance* between each fixation point inside a candidate AOI and the center of that AOI. The center coordinates (X_{center}, Y_{center}) of the AOI are calculated. The distance from Equation (6) and average distance using Equation (7) are then computed for each character with maximum fixation frequency using the formula:

$$\text{distance} = \sqrt{((\text{fixation}[0] - X_{center})^2 + (\text{fixation}[1] - Y_{center})^2)} \quad (6)$$

$$\text{avg_distance} = \frac{1}{n} \sum_{i=1}^n \text{distance} \quad (7)$$

where n is the number of fixation points within the AOI, and (X_i, Y_i) represents the coordinates of each fixation point. This step serves to disambiguate between characters that share the same fixation frequency by favoring the character where the fixations are more concentrated toward the center of the AOI.

From the filtered set of characters with maximum fixation frequency, the algorithm selects the character with the lowest average fixation distance to the center of its AOI. This character

is then designated as the final predicted output. This process effectively integrates fixation clustering, spatial analysis, and AOI-based mapping using data processing tools such as *Pandas* (for structured data manipulation) and *NumPy* (for numerical and distance computations). The use of predefined AOIs enables consistent mapping of gaze behavior across participants and supports a robust, data-driven method for interpreting user intent on the keyboard.

The proposed algorithm leverages fixation frequency and spatial distribution of gaze points to predict character selections in a novel eye-tracking-based keyboard system. The incorporation of AOI analysis ensures that visual attention is precisely mapped to specific keys, while the use of average distance refines the prediction by incorporating spatial coherence. This methodology contributes to an accessible and intelligent character selection mechanism tailored for individuals with ASD.

Algorithm 2 determines the most likely character selected by a user based on fixation data. It compares fixation coordinates against Areas of Interest (AOIs) defined by keyboard character boundaries, calculates fixation frequency per AOI, and refines prediction using average spatial distance between fixations and character centers. Using the `fixation_list` (L) containing gaze coordinates and the `keyboard_coordinates` (k) DataFrame with area-of-interest (AOI) information, the output is the `final_character` (ω) predicted based on where the user fixated. To enhance clarity, a flowchart has been included in Figure 9 to illustrate the process more effectively.

ALGORITHM 2 | Character selection.

CHARACTER-SELECTION (L, k, ω)

```

1. for each row in  $k$ 
2.   do row['fixation_frequency']  $\leftarrow 0$ 
3. for each fixation in  $L$ 
4.   do for each row in  $k$ 
5.     do if is_within_aoi(fixation, row)
6.       then row['fixation_frequency'] += 1
7. max_frequency  $\leftarrow$  maximum value in fixation_frequency column
8. max_frequency_chars  $\leftarrow$  all rows where fixation_frequency == max_frequency
9. for each row in max_frequency_chars do
10.   $X_{center} \leftarrow (row['X\_MIN'] + row['X\_MAX']) / 2$ 
11.   $Y_{center} \leftarrow (row['Y\_MIN'] + row['Y\_MAX']) / 2$ 
12.  total_distance  $\leftarrow 0$ 
13.  count  $\leftarrow 0$ 
14.  for each fixation in fixation_list
15.    do if is_within_aoi(fixation, row)
16.      then calculate distance
17.      total_distance += distance
18.      count += 1
19.  if count > 0
20.    then calculate avg_distance
21.    row['average_distance']  $\leftarrow$  avg_distance
22.  $\omega \leftarrow$  row in max_frequency_chars with minimum average_distance
23. return  $\omega$ 

```

3.8 | Proposed Optimized Ensemble Algorithm With a Meta-Learner

The ensemble learning strategy adopted in this research integrates predictions from three base classifiers such as Random Forest (RF), Support Vector Machine (SVM), and Neural Network (NN). Each base classifier produces a probability estimate for the predicted class based on extracted fixation features. These estimates are then used as meta-features in a logistic regression-based meta-learner to determine the final class prediction. The *prediction of the Random Forest model* is obtained by averaging the outputs of = 500 decision trees as Equation (8):

$$P_{RF}(x) = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (8)$$

where $h_t(x)$ denotes the class prediction from the t^{th} decision tree, and T is the total number of trees in the ensemble. This method improves robustness by reducing variance through ensemble averaging [39]. To enable the Support Vector Machine to perform non-linear classification, the input data is transformed using the *Radial Basis Function (RBF) kernel* Equation (9):

$$K(x_i, x) = e^{-\gamma \|x_i - x\|^2} \quad (9)$$

where γ is the kernel smoothing parameter, x_i represents a support vector, and x is the input feature vector. This transformation allows the classifier to learn more complex boundaries [40]. The *decision score* from the SVM is computed using following Equation (10):

$$f(x) = \sum_{i=1}^n \alpha_i K(x_i, x) + b \quad (10)$$

where α_i is the support vector coefficient, $K(x_i, x)$ is the RBF kernel function, and b is the learned bias. This function returns a real-valued score used for classification [41]. The *raw SVM decision score* is calibrated into a probability using a *logistic sigmoid function* Equation (11):

$$P_{SVM}(x) = \frac{1}{1 + e^{(Af(x)+B)}} \quad (11)$$

where A and B are calibration parameters obtained through [42]. The *feedforward neural network* used in this ensemble algorithm consists of multiple hidden layers, with architecture parameters set as *hidden_layer_sizes* = (200, 100) and *max_iter* = 500. The input is propagated forward through each layer using the following transformation (12):

$$z^{(l)} = \sigma(W^{(l)} h^{(l-1)} + b^{(l)}) \quad (12)$$

where $z^{(l)}$ is the pre-activation output of the l^{th} layer, $W^{(l)}$ is the weight matrix for layer l , $h^{(l-1)}$ is the activation from the previous layer, $b^{(l)}$ is the bias vector for layer l , σ is the activation function (ReLU) [43]. This operation is repeated for each layer in the network. The transformation maps input features into more abstract representations, allowing the network to learn complex non-linear relationships in the fixation feature space. The final hidden layer's output is passed to the softmax

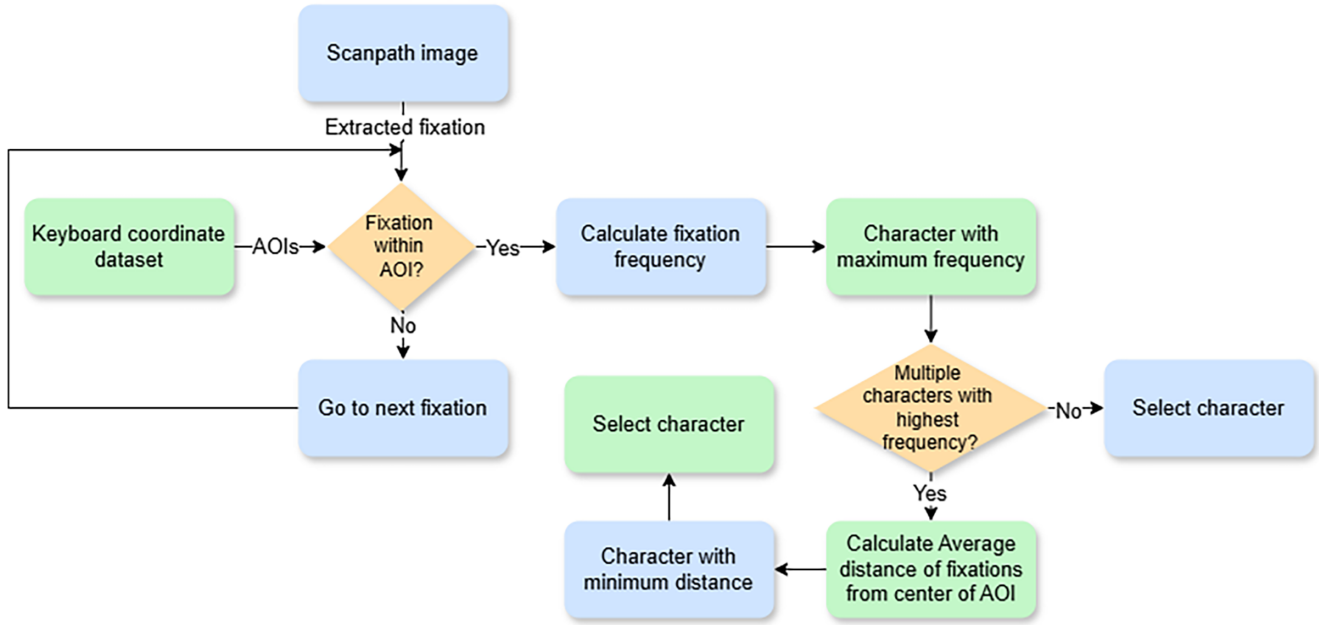


FIGURE 9 | Character selection process flowchart.

layer to compute class probabilities. The *Neural Network* outputs a class probability distribution using the *softmax function* Equation (13):

$$P_{NN}(x) = \frac{e^{z_j}}{\sum_{i=1}^C e^{z_i}} \quad (13)$$

Here, z_j is the logit for the predicted class, z_i are the logits for all C classes. This normalization ensures the output values represent valid probability estimates [43]. The final probability prediction is made using a *logistic regression meta-classifier* Equation (14) that takes the individual model predictions as input:

$$P(Y = 1|X_{meta}) = \frac{1}{1 + e^{-(\omega^T X_{meta} + b)}} \quad (14)$$

where $X_{meta} = [P_{RF}, P_{SVM}, P_{NN}]$ is the meta-feature vector, ω is the weight vector learned during logistic regression training, and b is the bias term. This meta-level prediction aggregates the strength of the three base classifiers (Hosmer [44]). The final predicted class $\hat{y}$ is chosen based on the maximum probability from the logistic regression output (Equation 15):

$$\hat{y} = \arg \max P(Y|X_{meta}) \quad (15)$$

This strategy ensures the ensemble algorithm selects the class with the highest confidence derived from the combined information across all base models.

The algorithm uses a custom ensemble approach, combining Random Forest (RF), Support Vector Machine (SVM), and Neural Network (NN) as base models, leveraging their individual strengths to improve decision-making. Furthermore, it employs an optimized meta-learner in the form of Logistic Regression, which stacks the predictions from the base models and enhances overall prediction accuracy. By creating meta-features from the

ALGORITHM 3 | Optimized ensemble.

OPTIMIZED-ENSEMBLE ($X, y, \hat{y}$)

1. **for** $t = 1$ to 500
2. Predict $h_t(x)$ from tree t
3. Calculate $P_{RF}(x)$
4. **for** $i = 1$ to n (support vectors)
5. Compute kernel similarity $K(x_i, x)$
6. Compute decision score $f(x)$
7. Compute probability $P_{SVM}(x)$
8. Pass x through fully connected layers with sizes (200, 100)
9. **for** each layer l
10. Calculate $z^{(l)}$
11. Compute logits z_j for each class
12. Compute softmax probability $P_{NN}(x)$
13. Create meta-feature vector $X_{meta} = [P_{RF}(x), P_{SVM}(x), P_{NN}(x)]$
14. Compute final probability $P(Y = 1|X_{meta})$
15. Select predicted class $\hat{y}$ using logistic regression meta-learner
16. **Return** $\hat{y}$ with highest predicted probability from logistic regression

base models' predictions, it develops a more discriminative and informative feature space for final classification. The algorithm is validated using a train-test split approach, and its performance is assessed through accuracy measurement, ensuring a thorough evaluation. This structured ensemble approach ensures a balanced trade-off between bias and variance, making the algorithm more resilient, interpretable, and effective compared to standalone algorithms. The ensemble classification algorithm is trained using the original dataset: features X and corresponding labels y ; it returns $\hat{y}$ with the highest predicted probability from logistic regression (Algorithm 3).

3.9 | Basic Workflow

The scanpath images were generated directly from the children's eye movements as they looked in different directions—up, down, left, and right—visually representing their gaze paths. These scanpath images were further analyzed to extract fixation coordinates (X, Y), fixation frequency, and average distance to fixation, ultimately forming a structured dataset for machine learning analysis. This data acquisition process was part of a broader workflow designed to develop an assistive communication system for individuals with Autism Spectrum Disorder (ASD). The workflow illustrated in Figure 10 began with the collection of 320 scanpath images, which were then pre-processed, cleaned, and normalized to ensure consistency and reliability. Exploratory Data Analysis (EDA) was conducted to identify patterns and relationships within the data, guiding feature selection and algorithm development. Data augmentation techniques were applied to address data imbalance and improve algorithm robustness.

A machine-learning algorithm was then trained using the processed dataset, leveraging fixation attributes to predict characters on a hands-free keyboard. The algorithm's performance was evaluated using accuracy, precision, and recall, followed by fine-tuning to optimize both its accuracy and generalization. This systematic approach integrated eye-tracking data and machine learning to create an efficient assistive system for individuals with ASD, enhancing communication by enabling character selection without manual input.

3.10 | Ensemble Algorithm and Dataset Overview

In this study, a customized dataset was developed by processing a pre-existing collection of scanpath images obtained from 15 children with Autism Spectrum Disorder (ASD). As secondary users of this image data, fixation points and gaze features were extracted using machine learning-based techniques to construct a fully de-identified structured dataset suitable for algorithm training and evaluation. Since no prior research has been conducted using this derived dataset, an ensemble algorithm has been proposed to analyze and interpret the data effectively. To evaluate its performance, a comparative analysis has been conducted against several traditional machine learning algorithms using the same dataset. The results have demonstrated the superiority of the ensemble approach in

enhancing predictive accuracy, highlighting its potential for improving assistive communication technologies. This study has established a foundational benchmark for future research leveraging this customized dataset.

4 | Results

The results highlight the algorithm's accuracy, precision, recall, and F1-score, along with detailed insights from precision-recall analyses, showcasing its strengths and areas for improvement.

4.1 | Evaluation Metrics

In classification problems, evaluating algorithm performance involves understanding how well the algorithm distinguishes between classes and predicts outcomes accurately.

4.1.1 | Accuracy

The Optimized Ensemble Algorithm achieved an impressive accuracy of 83.44% significantly outperforming the base models. This result demonstrates the algorithm's strong ability to correctly predict outcomes across the dataset using the following (Equation 16):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (16)$$

Here, TP is True Positives, FP is False Positives, TN is True Negatives, and FN is False Negatives.

4.1.2 | Macro-Averaged Precision

Precision measures the proportion of correctly predicted positive instances out of all positive predictions, calculated as Equation (17):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (17)$$

In multi-class classification, macro-averaging computes precision for each class and averages them equally. A precision 0.8143 means 81.43% of positive predictions are correct, reflecting the algorithm's accuracy in identifying true positives.

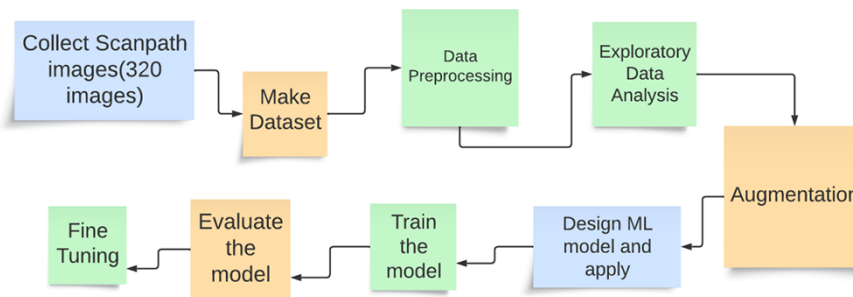


FIGURE 10 | System model workflow.

4.1.3 | Macro-Averaged Recall

Recall, or Sensitivity, measures the proportion of actual positive instances correctly identified by the algorithm, calculated as Equation (18):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (18)$$

A recall of 0.8057 means the algorithm correctly identifies 80.57% of actual positives, indicating some true positives are missed.

4.1.4 | Macro-Averaged F1-Score

The F1-score, the harmonic mean of precision and recall, balances false positives and false negatives, making it ideal for imbalanced datasets. It is calculated as the following Equation (19):

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (19)$$

In multi-class classification, it is averaged across all classes. A macro-averaged F1-score of 0.8043 reflects a good balance between identifying positives and minimizing errors.

4.2 | Diagnostic Metrics

Diagnostic metrics are used to gain deeper insights into the performance of a classification algorithm, particularly in terms of its ability to correctly identify different classes. These metrics, such as the confusion matrix, precision-recall curve, and AUC, help in understanding how well the algorithm distinguishes between classes, revealing areas where it may need improvement, especially in the case of class imbalance or misclassification patterns.

4.2.1 | Precision-Recall Curve for Multi-Class Classification

The evaluation of the algorithm's performance using the *Precision-Recall (PR) curve* in Figure 11 demonstrates its robustness and reliability in multi-class classification. A key highlight is the algorithm's ability to achieve perfect Average Precision (AP) scores of 1.00 for multiple classes, including ENTER, I, and Q, indicating that these classes are classified with no false positives or false negatives. Furthermore, several other classes, such as A, D, G, N, S, T, V, Y, and Z, exhibit AP scores above 0.90, reinforcing the algorithm's capability to maintain both high precision and recall. This result suggests that the algorithm can effectively differentiate between these classes, ensuring that relevant instances are retrieved while minimizing incorrect classifications.

The PR curve structure further supports the algorithm's strength in precision, where certain classes maintain precision values close to 1.0 across the entire recall range. This stability in precision implies that the algorithm confidently makes correct predictions with minimal misclassification. Additionally, the algorithm exhibits a well-balanced trade-off between precision and recall, ensuring not only that correct instances are identified but also that very few relevant instances are missed. This balance is particularly important in real-world applications, where both precision (reducing false positives) and recall (minimizing false negatives) are critical for optimal classification outcomes.

Despite the algorithm's overall strong performance, certain classes demonstrate lower AP scores that warrant further investigation. Notably, Class K (AP = 0.17) performs significantly below the others, suggesting difficulties in distinguishing it from other categories. This could be attributed to class imbalance, feature similarities with other classes, or limited representation in the training data. Similarly, Class M (AP = 0.69) and Class W (AP = 0.74)

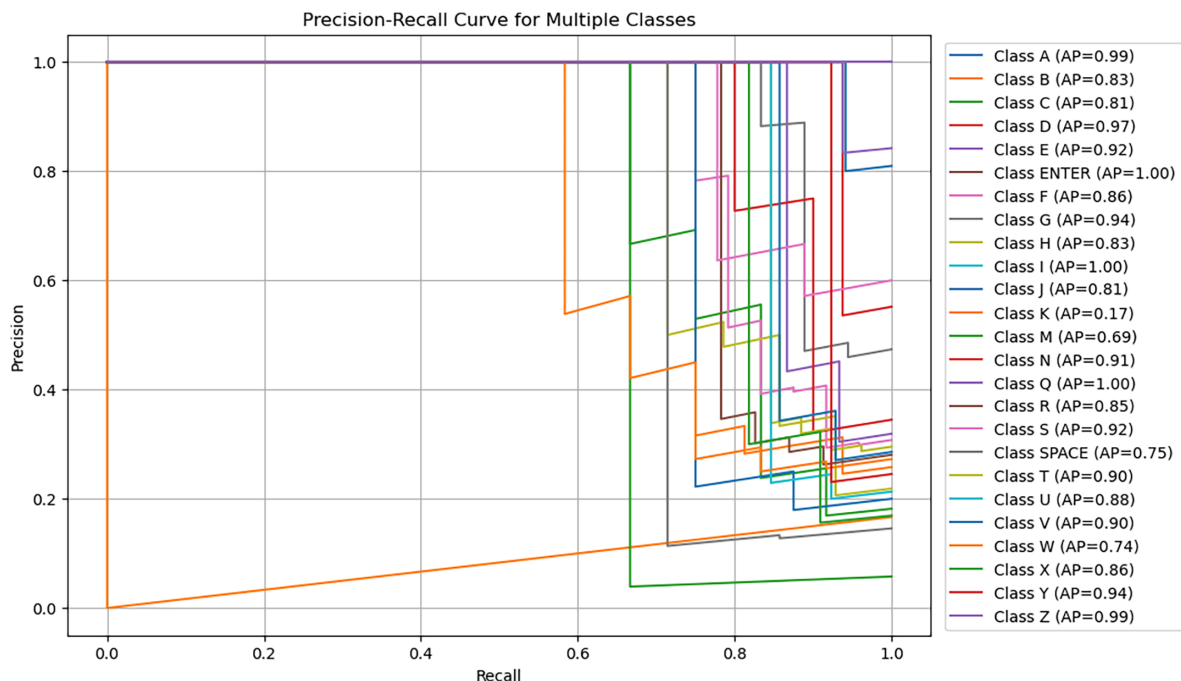


FIGURE 11 | Precision-recall curve using proposed ensemble algorithm.

(AP = 0.74) fall slightly below the desirable threshold, indicating potential areas for fine-tuning and optimization. Addressing these issues through data augmentation, feature extraction, or adjusting class weighting strategies during training could help improve classification performance for these classes.

Overall, the high AP scores across the majority of classes demonstrate that the algorithm is highly effective at multi-class classification, achieving remarkable precision and recall. The presence of multiple classes with near-perfect classification further validates its robustness and accuracy. While a few classes exhibit lower performance, targeted refinements in data representation and algorithm training strategies could significantly enhance these areas. With these improvements, the algorithm has the potential to achieve even greater classification accuracy and generalization capability, making it a reliable and well-calibrated multi-class classifier.

4.2.2 | Area Under the Precision-Recall Curve

The *Area Under the Precision-Recall curve (AUC-PR)* is a crucial metric for evaluating the performance of a classification algorithm, particularly in imbalanced datasets. In Figure 11, each colored line represents a Precision-Recall (PR) curve for a different class, and the AP (Average Precision) values in the legend correspond to the AUC-PR for each class. A higher AUC-PR indicates better classification performance. Several classes, such as ENTER, I, and Q, have an AP score of 1.00, meaning their AUC-PR is also 1.00 (100%), signifying perfect classification with zero false positives and zero false negatives. Additionally, classes with AP scores close to 1.00, such as A (0.99), D (0.97), G (0.94), and Z (0.99), perform exceptionally well, maintaining high precision and recall across the dataset. However, some classes exhibit lower AUC-PR values, indicating difficulty in classification. For example, Class K (AP = 0.17) has the lowest AUC-PR, suggesting that the algorithm struggles to distinguish this class correctly, likely due to low precision and recall values, leading to a high number of misclassifications. Similarly, Class M (AP = 0.69) and Class W (AP = 0.74) show moderate performance, meaning that while the algorithm detects them reasonably well, further improvements could enhance accuracy. AUC-PR is a more relevant metric than ROC-AUC in imbalanced classification tasks since it focuses on false positives and false negatives rather than true negatives. A high AUC-PR value indicates that the algorithm is effective in retrieving relevant samples while keeping false positives low. Overall, this algorithm demonstrates exceptionally high performance across most classes, with several achieving near-perfect classification. However, certain classes (K, M, and W) require additional improvements, which could be addressed through data augmentation, class balancing, or enhanced feature extraction. The overall high AUC-PR scores suggest that this algorithm is well-calibrated and highly effective in multi-class classification tasks.

5 | Comparative Analysis

The proposed ensemble algorithm exhibits enhanced performance relative to conventional classifiers such as Multilayer Perceptron (MLP), Random Forest, and Support Vector Machine

(SVM). This improvement can be attributed to the ensemble's capacity to integrate multiple weak learners, thereby mitigating both bias and variance. Furthermore, the algorithm effectively addresses challenges related to class imbalance, contributing to greater robustness and generalizability. Unlike individual classifiers, which are often prone to overfitting and noise sensitivity, the ensemble approach offers a more stable and reliable framework for classification tasks.

5.1 | Algorithm Evaluation, Validation and Augmentation Strategy

A comprehensive evaluation of the proposed Optimized Ensemble Approach was conducted using a wide range of performance metrics beyond conventional accuracy. The dataset was normalized and augmented to mitigate class imbalance, followed by a stratified 80/20 train-test split to maintain representative distributions during training and evaluation.

The dataset used for character-level prediction was constructed from AOI-based fixation analysis, where each label was algorithmically derived using fixation frequency and the average distance from the AOI. However, these AOI-derived labels contained noise and exhibited significant class imbalance, posing challenges for algorithm generalization. The augmentation function was applied exclusively to the training split (X_{train}, y_{train}) after stratified, while the test set remained unchanged. Prior to augmentation, the training set contained 286 samples and the test set contained 32 samples. Following augmentation, the training set increased to 1144 samples, whereas the test set size remained constant at 32 samples shown in Figure 12. This demonstrates that no augmented samples were introduced into the test set, thereby eliminating any possibility of information leakage and ensuring the validity of the evaluation. To prevent train-test leakage, a k-fold cross-validation strategy was also applied. While this approach ensured strict isolation between subjects in training and testing sets, it substantially reduced the effective number of samples per fold. Consequently, the algorithm encountered difficulties in learning discriminative patterns, leading to a noticeable decline in predictive performance, with classification accuracy dropping to 18.75%.

To address these challenges, an ensemble-based approach was explored by integrating minority-class oversampling and train-set augmentation while maintaining a strict separation between training and testing samples. By leveraging augmented AOI features, the proposed method demonstrated improved generalization capability under the experimental setup. In these experiments, an overall accuracy of 83.44% was achieved, along with relatively higher precision and recall across several character classes.

The effect of the validation strategies is illustrated by the Precision-Recall (PR) curves presented in Figures 11 and 13. As shown in Figure 13, the k-fold cross-validation approach produces PR curves with comparatively lower area under the curve (AUC) for most classes, indicating limited discriminative ability. In contrast, Figure 11 shows that the ensemble-based approach yields higher precision and recall across multiple classes under the same dataset conditions.

Train shape: (286, 62) (286,)
 Test shape : (32, 62) (32,)
 Final Train shape after subtle augmentation: (1144, 62) (1144,)

FIGURE 12 | Verification of training-only augmentation: Dataset sizes before and after augmentation.

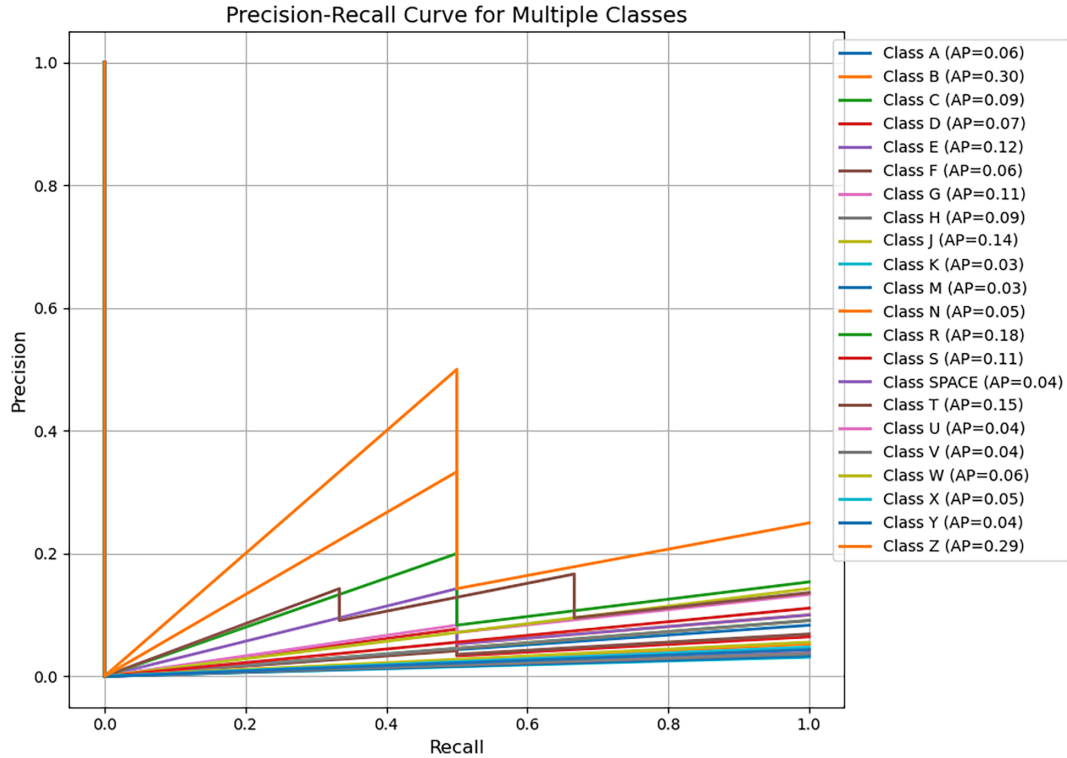


FIGURE 13 | Precision-recall curve using k-fold cross validation method.

In summary, the k-fold cross-validation strategy effectively prevents train-test leakage but may lead to under-fitting due to reduced sample diversity per fold. Given the noisy and imbalanced nature of AOI-derived labels, the experimental findings suggest that augmentation-aware ensemble strategies can help extract useful patterns and achieve better predictive performance.

For classification assessment, macro-averaged precision, recall, and F1-score were calculated. These metrics are particularly suitable for multi-class classification tasks, as they assign equal weight to each class regardless of frequency, thereby reducing potential bias introduced by class imbalance. Additionally, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) were measured to assess the magnitude and relative scale of prediction deviations. Statistical measures such as variance, standard deviation, and the coefficient of variation (CV) were computed to analyze the consistency and dispersion of predictions, while skewness was examined to explore potential asymmetries in the error distribution.

5.2 | Error and Variability Analysis in Algorithm Performance

The error metrics show that the ensemble algorithm consistently outperforms individual algorithms such as MLP, Random Forest,

and SVM, achieving lower error rates and higher stability. By effectively handling class imbalance and reducing variance, the ensemble approach proves to be the most reliable and accurate for eye-gaze-based character selection.

5.2.1 | Mean Squared Error (MSE)

The MSE (Mean Squared Error) as defined in Equation (20) measures the average squared difference between actual and predicted values. A lower MSE indicates that the algorithm's predictions are closer to the actual values, reducing the impact of large errors.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - y'_i)^2 \quad (20)$$

where y_i is the actual value, y'_i is the predicted value, n is the total number of predictions.

The Optimized Ensemble Algorithm significantly outperforms all other models with the lowest MSE (20.5682). In contrast, MLP has the highest MSE (57.5831), indicating that its predictions have larger errors. The lower MSE in the ensemble algorithm suggests that it generalizes better across different classes, even with data imbalance. This improvement is likely due to the combination of multiple weak learners, leading to a more robust algorithm.

5.2.2 | Root Mean Squared Error (RMSE)

The RMSE (Root Mean Squared Error) defined in Equation (21) provides an interpretable measure of error in the same unit as the target variable. It penalizes larger errors more than smaller ones, making it a crucial metric for assessing prediction quality.

$$RMSE = \sqrt{MSE} \quad (21)$$

The Optimized Ensemble Algorithm achieves the lowest RMSE (4.5352), reinforcing that it has the most precise predictions among all algorithms. MLP has the highest RMSE (7.5884), indicating poor prediction accuracy. The ensemble algorithm's success in lowering RMSE confirms that it effectively reduces large errors while handling class imbalance better.

5.2.3 | Mean Absolute Percentage Error (MAPE)

MAPE (Mean Absolute Percentage Error) defined in Equation (22) calculates the error as a percentage of actual values, making it useful for understanding relative prediction errors. A lower MAPE indicates better prediction performance. The formula for MAPE is:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - y'_i}{y_i} \right| \quad (22)$$

where: y_i is the actual value of the i^{th} observation, y'_i is the predicted value of the i^{th} observation, n is the total number of observations.

The SVM model has the lowest MAPE (41.9792%), meaning it provides the most accurate percentage-based error reduction. The ensemble algorithm (43.8389%) also performs well, demonstrating its robustness. However, MLP has the worst MAPE (67.34%), meaning its predictions significantly deviate from actual values. The ensemble algorithm's balanced performance suggests that it learns well across different classes despite the imbalance.

5.2.4 | Variance

The Variance defined in Equation (23) measures the spread of errors. Lower variance means a more stable and reliable algorithm, while high variance suggests inconsistent predictions. The formula for Variance is:

$$Variance = \frac{1}{n-1} \sum_{i=1}^n (y_i - y'_i)^2 \quad (23)$$

The formula for y'_i is:

$$Sample\ mean = \frac{1}{n} \sum_{i=1}^n y_i \quad (24)$$

where y_i is the individual data point, y'_i is the sample mean defined in Equation (24), n is the sample size.

The ensemble algorithm has the lowest variance (20.6263), making it the most consistent in its predictions. High variance in MLP (57.7706) indicates erratic predictions, likely due to overfitting. The ensemble approach reduces variance by combining diverse models, thus avoiding overfitting.

5.2.5 | Standard Deviation (STD)

Standard Deviation (STD) defined in Equation (25) represents the dispersion of errors. Lower STD means a more stable algorithm with consistent prediction performance. The formula is:

$$STD = \sqrt{Variance} \quad (25)$$

The ensemble algorithm has the lowest STD (4.5352), making it the most stable. The MLP model has the highest STD (7.6007), showing that its predictions are highly variable. By combining multiple classifiers, the ensemble algorithm smooths out extreme variations, producing more reliable results.

5.2.6 | Coefficient of Variation (CV)

CV (Coefficient of Variation) defined in Equation (26) measures error variation relative to the mean. A higher CV indicates more variation in predictions. The formula is:

$$CV = \frac{STD}{Sample\ mean} \quad (26)$$

The ensemble algorithm has the highest CV (2.8723), meaning its errors are more spread out compared to other models. However, since its overall performance metrics are superior, this does not negatively impact the algorithm.

5.2.7 | Skewness

Skewness defined in Equation (27) measures the symmetry of errors. A negative skew indicates under predictions, while a positive skew suggests over predictions. The formula is:

$$Skewness = \frac{\sum (y_i - y'_i)^3}{(n-1) \cdot (STD^3)} \quad (27)$$

The ensemble algorithm has a low negative skew (−0.2532), indicating balanced errors. Random Forest has the highest skew (−0.8850), meaning it under predicts significantly. The ensemble algorithm remains unbiased and stable across classes.

The Optimized Ensemble Algorithm outperforms traditional models in almost all metrics. It has the lowest MSE, RMSE, variance, and standard deviation, proving it is the most accurate and stable. Even with an imbalanced multiclass dataset, it effectively generalizes well across classes. The improvement is due to ensemble learning, which combines multiple models to reduce bias and variance, making it more robust than individual models like MLP, Random Forest, and SVM.

5.3 | Comparison Between Baseline Model and Proposed Algorithm

The Table 2 compares the classification accuracy of MLP, Random Forest, SVM, and an Optimized Ensemble Algorithm. MLP achieved the lowest accuracy at 62.66%, likely due to its sensitivity to noise and limited generalization on this dataset. Random

TABLE 2 | Performance metrics of machine learning algorithms for eye-gaze-based keyboard selection.

Algorithm	Accuracy	MSE	RMSE	MAPE	Variance	STD	CV	Skewness
MLP	62.66%	57.5831	7.5884	67.3476%	57.7706	7.6007	1.9741	−0.1037
Random Forest	75.89%	47.7305	6.9087	45.3384%	47.5648	6.9087	2.1902	−0.8850
SVM	76.60%	38.5248	6.2068	41.9792%	38.5925	6.2068	2.5612	−0.4598
Optimized Ensemble Algorithm	83.44%	20.5682	4.5352	43.8389%	20.6263	4.5352	2.8723	−0.2632

Forest showed a significant improvement with 75.89%, benefiting from its ability to handle non-linear relationships and reduce overfitting. SVM slightly outperformed Random Forest with 76.60%, likely due to its margin-maximizing approach, which worked well for the dataset's feature distribution. The Optimized Ensemble Algorithm achieved the highest accuracy at 83.44% by combining predictions from the individual models, leveraging their complementary strengths, and reducing biases. This highlights the ensemble algorithm's superiority in achieving robust and accurate classification, demonstrating its ability to outperform standalone models. The Optimized Ensemble Algorithm consistently outperforms the other models across several metrics, with the lowest Mean Squared Error (MSE) of 20.5682 and Root Mean Squared Error (RMSE) of 4.5352, indicating superior algorithm performance. It also has the lowest Mean Absolute Percentage Error (MAPE) of 43.3389%, demonstrating its efficiency in predicting values. In terms of variance and standard deviation (STD), the Optimized Ensemble Algorithm also shows the least variation in its predictions (Variance: 20.6263, STD: 4.5352). Furthermore, it achieves a relatively lower coefficient of variation (CV) of 2.8723 and a skewness of −0.2632, indicating a more balanced and stable prediction distribution compared to MLP, Random Forest, and SVM.

6 | Conclusion

In conclusion, this research developed an educational assistive system tailored for individuals with Autism Spectrum Disorder (ASD), integrating eye-tracking technology and machine learning algorithms into a smart board. By leveraging fixation values and Area of Interest (AOI) analysis, the system introduced a novel, hands-free keyboard selection method intended to support communication and learning experiences. The evaluation using an Optimized Ensemble Algorithm showed strong performance, achieving an accuracy of 83.44% and outperforming baseline models, thereby supporting the system's functionality. Utilizing a pre-existing collection of scanpath images from 15 Bangladeshi children with ASD, gaze-related features were extracted and processed to create a customized dataset for the machine learning pipeline. This work demonstrates technical effectiveness and offers a culturally relevant, dataset-driven reference for future assistive technologies in underrepresented communities. The proposed system shows potential to foster better interaction with parents, teachers, and caregivers, which may contribute to promoting social engagement and improving quality of life for neurodiverse learners. Overall, this research underscores the potential role of innovative technologies in creating inclusive and effective educational solutions for individuals with ASD.

Author Contributions

Sadia Afrin: conceptualization, methodology, software, data curation, investigation, writing – original draft, writing – review and editing, visualization, formal analysis, resources. **Afrin Ahmed:** conceptualization, formal analysis. **Mehrin Anannya:** supervision. **Samrat Kumar Dey:** investigation, validation, visualization. **Md. Moshir Rahman:** supervision. **Rashed Mazumder:** project administration, supervision.

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Ethics Statement

This study uses a secondary dataset originally collected with appropriate ethical approval that can be accessed at this link: [AssistiveAI-Autism/Data Permission at main AfrinSadi/AssistiveAI-Autism GitHub](#). All procedures performed in studies involving human participants were in accordance with the ethical standards of the Biosafety, Biosecurity, and Ethical Clearance Committee of Jahangirnagar University, Savar, 1342 Dhaka, Bangladesh and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent

Informed consent was obtained from all individual participants included in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in AssistiveAI-Autism at <https://github.com/AfrinSadi/AssistiveAI-Autism.git>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.